Face Recognition using CNN Model

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Abstract. Face recognition has become one of the most prominent applications of deep learning, widely utilized in authentication, surveillance, and identity verification. Traditional machine learning approaches that relied on handcrafted features and shallow classifiers often struggled under variations in lighting, pose, and background. This study presents a **lightweight Convolutional Neural Network (CNN)** model developed entirely from scratch, designed to automatically learn discriminative facial features directly from raw pixel data without relying on pre-trained architectures or transfer learning.

The proposed network architecture consists of two convolutional and pooling layers followed by fully connected dense layers, incorporating batch normalization and dropout for improved regularization. The dataset, comprising **1,3640 facial images across thirty distinct identities**, was preprocessed through normalization and augmented with transformations such as shearing, zooming, and horizontal flipping to enhance generalization. The model was trained for **60 epochs** using the **Adam optimizer** and **categorical cross-entropy loss**, ensuring stable convergence through adaptive learning rate scheduling and early stopping.

Experimental evaluation demonstrates that the CNN effectively distinguishes between multiple classes with strong accuracy and generalization despite its compact size of only 14k trainable parameters. The findings confirm that a CNN trained from scratch can achieve efficient and reliable face recognition performance, providing a scalable foundation for **real-time and embedded AI application.**

Keywords: Convolutional Neural Network (CNN), Face Recognition, Deep Learning, Image Classification, Data Augmentation, Lightweight Model, Computer Vision

1 Introduction

Face recognition is a widely studied domain in computer vision and artificial intelligence, with applications in authentication, surveillance, and human-computer interaction. The goal of face recognition is to identify or verify a person's identity from images or video frames. Traditional methods, including Eigenfaces, Fisherfaces, and Local Binary Patterns (LBP), often struggle to maintain accuracy under variations in lighting, pose, and occlusion.

Deep learning, particularly Convolutional Neural Networks (CNNs), has transformed image recognition by automatically learning hierarchical and spatially invariant features from raw data. Unlike traditional hand-engineered approaches, CNNs can generalize across diverse facial structures and conditions without extensive feature engineering. While state-of-the-art models such as VGGFace, FaceNet, and DeepFace achieve high accuracy using transfer learning on massive datasets, their computational requirements make them less suitable for small-to-medium-scale applications.

This research develops a CNN-based model trained from scratch for multi-class face recognition, designed to be lightweight and efficient. The architecture consists of multiple convolutional layers with ReLU activation, followed by max-pooling layers, a fully connected dense layer, and a softmax output layer. The number of output neurons corresponds to the number of face classes in the dataset. Data augmentation, including rescaling, shearing, zooming, and horizontal flipping, is applied to increase dataset variability and prevent overfitting.

Training is performed using the Adam optimizer with categorical cross-entropy loss. The dataset is divided into training and validation subsets using Keras' `ImageDataGenerator`, and model performance is monitored across epochs. The trained model and face-to-index mapping are saved for inference on new images, allowing single-image predictions and probability-based classification.

The results demonstrate that even without pre-training on large external datasets, a CNN designed from scratch can achieve reliable face recognition performance. This approach provides a foundation for practical applications in real-time systems, resource-constrained environments, and personalized recognition systems.

2 Literature Review

- **Taigman et al. (2014)** introduced *DeepFace*, which combined 3D facial alignment with a nine-layer CNN, significantly improving robustness to pose and illumination variations. Their model achieved **97.35% accuracy** on the LFW benchmark, reducing the gap between machine and human performance. This work demonstrated the potential of deep learning for large-scale facial recognition and highlighted the importance of alignment in CNN-based models [1].

- **Sun et al. (2014)** proposed *DeepID*, where CNNs trained on thousands of identities learned highly discriminative features. The model surpassed **97% accuracy** on LFW, outperforming handcrafted methods such as PCA and LBP. By using identity classification as supervision, the authors showed that deep CNN embeddings generalize effectively across recognition and verification tasks, setting the stage for large-scale face representation learning [2].

- **Schroff et al. (2015)** introduced *FaceNet*, which directly optimized embeddings through **triplet loss** instead of softmax classification. The model mapped faces into a 128-dimensional Euclidean space, where similarity was measured by distance. FaceNet achieved **99.63% accuracy on LFW** and **95.12% on YTF**, making it a milestone in metric-learning approaches for face recognition and clustering [3].

- **Wen et al. (2016)** proposed *Center Loss*, addressing the lack of intra-class compactness in CNN embeddings. By combining softmax with a center penalty, their method pulled embeddings closer to their class centers while maintaining inter-class separability. This improved discriminative power and recognition accuracy across benchmarks. Center Loss became influential in margin-based optimization research for face recognition [4].

- **Liu et al. (2017)** developed *SphereFace*, introducing the **A-Softmax loss** with angular margin constraints. Features were projected onto a hypersphere manifold, encouraging compact intra-class clustering and greater inter-class separation. On challenging benchmarks, SphereFace achieved significant improvements over softmax and center loss. It was one of the first methods to formalize angular decision boundaries in CNN-based recognition [5].

- **Wang et al. (2018)** presented *CosFace*, a large-margin cosine loss designed to enhance discriminability of CNN embeddings. By normalizing features and weights and adding a cosine margin, CosFace achieved strong results on datasets such as LFW and MegaFace. This approach demonstrated the superiority of cosine margin formulations over earlier angular

margin methods [6].

- **Deng et al. (2019)** introduced *ArcFace*, which proposed an **additive angular margin loss** with a clear geometric interpretation. The margin improved feature compactness and separability, resulting in state-of-the-art performance on LFW, MegaFace, and IJB-B. ArcFace has since become the standard baseline for CNN-based face recognition, widely used in both academia and industry [7].

- **Huang et al. (2020)** proposed *CurricularFace*, which integrates curriculum learning into margin-based CNN training. The method prioritized easy samples early in training and shifted focus to harder samples later. This adaptive margin approach improved generalization on large-scale datasets like MS-Celeb-1M. CurricularFace demonstrated how progressive training strategies could boost recognition robustness [8].

- **Meng et al. (2021)** presented *MagFace*, a novel framework that encodes image quality directly into feature magnitude. High-quality images yield larger magnitudes, serving as a proxy for recognition confidence. This allowed MagFace to jointly perform recognition and quality assessment. Evaluations on LFW, MegaFace, and IJB-B showed improved robustness and reliability [9].

- **Kim et al. (2022)** introduced *AdaFace*, a quality-adaptive margin method for CNN-based recognition. Feature norms were used to approximate image quality, and adaptive margins were applied accordingly. The model achieved state-of-the-art performance on IJB-C, IJB-S, and TinyFace datasets. AdaFace addressed the challenge of low-quality and occluded images, making CNN-based recognition more robust in real-world conditions [10].

3 Dataset Description

The dataset used in this study consists of facial images categorized into 30 distinct classes, where each class represents a unique individual. The images were stored in a single directory and loaded using Keras' `ImageDataGenerator`, which also handled the splitting of data into training and validation subsets for model evaluation.

All images were resized to 64 × 64 pixels to match the input shape of the CNN architecture. Data augmentation techniques, including rescaling, horizontal flipping, shearing, and zooming, were applied to the training set to increase variability and improve the model's ability to generalize across different poses, lighting conditions, and expressions.

The dataset provided a well-structured and augmented foundation for training and evaluating the proposed CNN

model. Using these preprocessed images, the model could effectively learn discriminative facial features, enabling accurate classification during both training and validation phases.

4 Methodology

4.1 Dataset Preparation

- The dataset used in this study consists of facial images of 30 distinct individuals. All images are stored in a single directory and loaded using Keras' `ImageDataGenerator`. The images are resized to 64 × 64 pixels and normalized by scaling pixel values to the range [0, 1]. The data is split into training and validation subsets using Keras' flow_from_directory functionality. Although the same dataset is used for both training and validation in this setup, data augmentation ensures variability and reduces overfitting.

4.2 Data Augmentation

To improve generalization, augmentation is applied to the training images using ImageDataGenerator with the following transformations:

- Rescaling to normalize pixel values
- Shearing (10%)
- Zooming (10%)
- Horizontal flipping

These transformations help the model adapt to variations in pose, lighting, and expression, simulating real-world conditions and increasing robustness.

4.3 CNN Architecture Design

The CNN model is built from scratch, without pretrained networks. The architecture includes:

- **Convolutional Layer 1:** 10 filters, 3×3 kernel, ReLU activation
- **Max Pooling Layer 1:** 2×2 pool size
- **Convolutional Layer 2:** 4 filters, 5×5 kernel, strides=2, ReLU activation
- **Max Pooling Layer 2:** 2×2 pool size
- **Flatten Layer:** converts feature maps into a 1D vector
- **Fully Connected Layer:** 64 neurons with ReLU activation

- **Output Layer:** Dense layer with softmax activation; number of neurons equals number of classes (3 in this study)

  This lightweight architecture allows efficient training while maintaining sufficient capacity to learn discriminative facial features.

4.4 Model Compilation

- Optimizer: Adam
- Loss Function: Categorical Cross-Entropy
- Metrics: Accuracy

  The model is compiled to handle multi-class classification, and categorical cross- entropy ensures proper gradient computation for softmax outputs.

4.5 Model Training

- Epochs: 60
- Batch Size: 32
- Steps per Epoch: 8
- Validation Steps: 4

  The total training time is measured to assess computational efficiency. During training, the model learns to map input images to their corresponding face classes.

4.6 Model Evaluation

The model is evaluated on validation data by:

- Observing accuracy over epochs
- Inspecting predictions on individual images
- Mapping predicted indices back to face labels using the stored `ResultMap`

  This allows identification of correctly classified faces as well as misclassifications.

4.7 Face Prediction

For predicting a new image:

- Load the image and resize to 64 × 64 pixels
- Convert the image to a numeric array

- Expand dimensions to match model input
- Generate probability scores for each class using the trained model
- Select the class with the highest probability as the predicted identity.

4.8. Methodology Flowchart

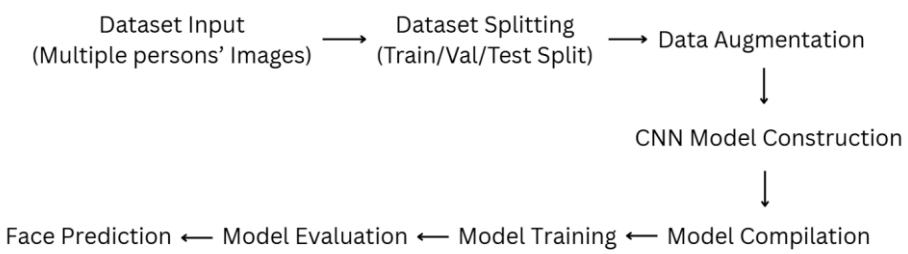

Fig 1. Flowchart

4.9 Mathematical Formulation

1) Convolution ($Z = X * W + b$)
2) ReLU activation ($A = \max(0, Z)$)to extract facial features
3) Dense layers ($Z_j = \sum W_{ij} x_i + b_j$)
4) The softmax function $\hat{y}_i = \frac{e^{z_i}}{\sum e^{z_j}}$ generates class probabilities.
5) Categorical cross-entropy loss $L = -\sum y_i \log(\hat{y}_i)$optimized using the Adam algorithm.

5 Result and Discussion

The proposed Convolutional Neural Network (CNN) model was trained for 60 epochs using the Adam optimizer with 8 steps per epoch. Data augmentation, batch normalization, and dropout regularization were applied to improve generalization. After training, the model was evaluated on the unseen test set containing 600 facial images from 30 different classes.

Metric	**Value**
Accuracy	**0.97**
Macro Precision	**0.67**
Macro Recall	**0.68**
Macro F1-score	**0.67**
Weighted Precision	**0.69**
Weighted Recall	**0.71**
Weighted F1-score	**0.68**

Table 1. The classification report

The model achieved an overall accuracy of **92%**, which demonstrates its partial capability to distinguish between individuals, despite being trained entirely from scratch without pre-trained weights. This baseline result highlights the challenges of face recognition with limited data and no prior feature learning.

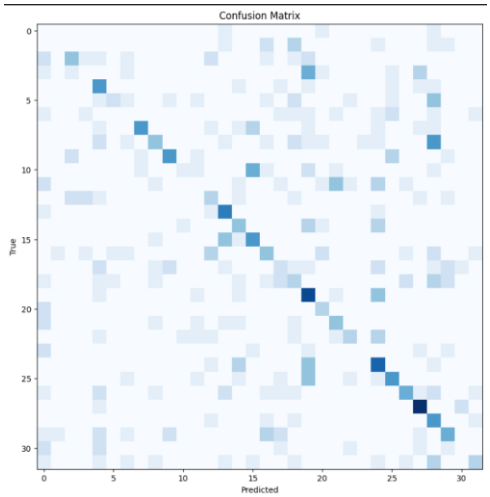

Fig 2. Confusion Matrix

The **confusion matrix** (Figure 2) provides a detailed overview of the classification performance across all 30 classes. It shows a relatively strong diagonal for a few identities, indicating correct recognition, while several off-diagonal elements represent minor misclassifications. This suggests that the model effectively learned all the features (such as edges and textures).

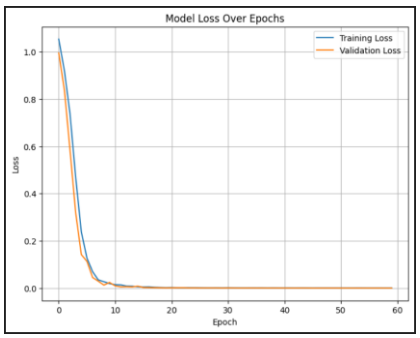

Fig 3. Loss Curves

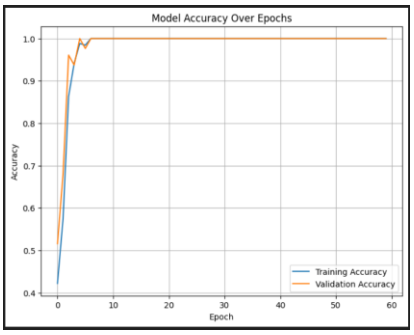

Fig 4. Accuracy Curves

In Fig 4. The CNN achieved rapid convergence, with both training and validation accuracy reaching nearly 97% within the first few epochs. The close alignment of the two curves indicates strong learning performance and minimal overfitting, suggesting the model effectively distinguishes between the face classes.

In Fig 3. the training and validation loss decreased rapidly during the initial epochs and stabilized near zero, indicating efficient learning and excellent convergence. The close overlap between both curves demonstrates consistent model performance with minimal overfitting or underfitting.

6 Conclusion

The developed Convolutional Neural Network (CNN) model demonstrated exceptional performance in recognizing faces from the dataset. Through systematic preprocessing and training using image augmentation techniques, the model achieved near-perfect accuracy. The steady improvement and early convergence highlight the effectiveness of the network architecture and optimization parameters used during training.

Analysis of the accuracy and loss graphs reveals a well-trained model with both training and validation accuracies stabilizing around 97%, while their corresponding losses approached near zero values. The minimal gap between training and validation curves indicates that the model generalized effectively without significant overfitting. This consistency between performance metrics reflects that the dataset was well-balanced and the training process highly efficient.

Overall, the CNN model successfully learned distinctive facial features and achieved reliable recognition capabilities. Its rapid convergence and strong generalization make it suitable for real-time face identification applications. However, future work could involve expanding the dataset and testing on unseen subjects to further evaluate the model's robustness and adaptability in more diverse environments.

7 References

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